

Bayesian Write Up

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Workflow write up: Note: Will need to edit to correspond to correct results and statistical analysis sections and hide code chunks

In this Bayesian inference, the beta binomial model will be used. T represents the number of infections with a distribution of:

$$T \mid \pi \sim \text{Binom}(170, \pi)$$

where π is:

$$\pi \sim \text{Beta}(a, b)$$

Our parameter of interest is vaccine efficacy represented by:

$$\psi = \frac{1 - 2\pi}{1 - \pi}$$

To determine a prior, an uninformative uniform prior of $\text{Beta} \sim (1, 1)$ can be used. However, we are choosing to use prior values based on assumptions on vaccine efficiency. So, the assumptions that we initially, with efficacy being assumed as unlikely, set were that the probability of the vaccine being at least 30% effective is 0.01. The probability that the vaccine has any effect is greater than 0 is 0.5. These can be later tested. Mathematically:

$$P(\psi \geq 0.3) = 0.01$$

$$P(\psi \geq 0) = 0.5$$

The probability statements in terms of π , since that is the variable we have a distribution for, is:

$$P(\psi = \frac{1 - 2\pi}{1 - \pi} \geq 0.3) = 0.01$$

$$P(\pi \leq \frac{7}{17}) = 0.01$$

$$P(\psi = \frac{1 - 2\pi}{1 - \pi} \geq 0) = 0.5$$

$$P(\pi \leq \frac{1}{2}) = 0.5$$

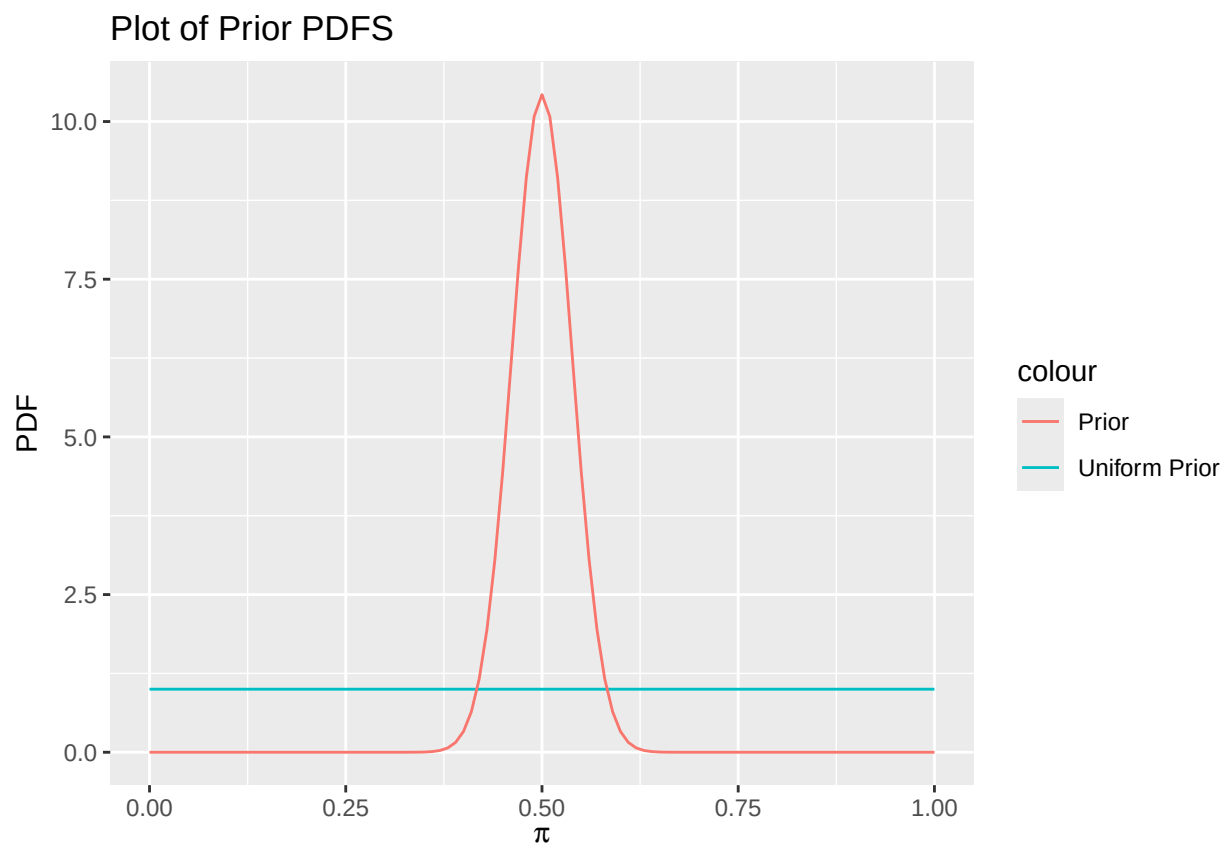
This means that the $\text{Beta}(a, b)$ prior for π is one with a median of 0.5 and 1st percentile of $\frac{7}{17}$:

```
beta.select(quantile1 = list(p = 0.01, x = 7/17),  
            quantile2 = list(p = 0.5, x = 0.5))
```

```
## [1] 85.63 85.63
```

The distribution of this is:

```
ggplot() +
  geom_function(fun = dbeta,
    mapping = aes(color = "Uniform Prior"),
    args = list(shape1 = 1, shape2 = 1),
    xlim = c(0, 1)) +
  geom_function(fun = dbeta,
    mapping = aes(color = "Prior"),
    args = list(shape1 = 85.63, shape2 = 85.63),
    xlim = c(0, 1)) +
  labs(
    title = "Plot of Prior PDFS",
    x = expression(pi),
    y = "PDF")
```



As stated above, the prior distribution is $\pi \sim \text{Beta}(85.63, 85.63)$. To determine a posterior distribution, the shape parameters are $a + t$ and $b + n - t$, with t being the observed value for T . In this case, $n = 170$ and $t = 8$. So:

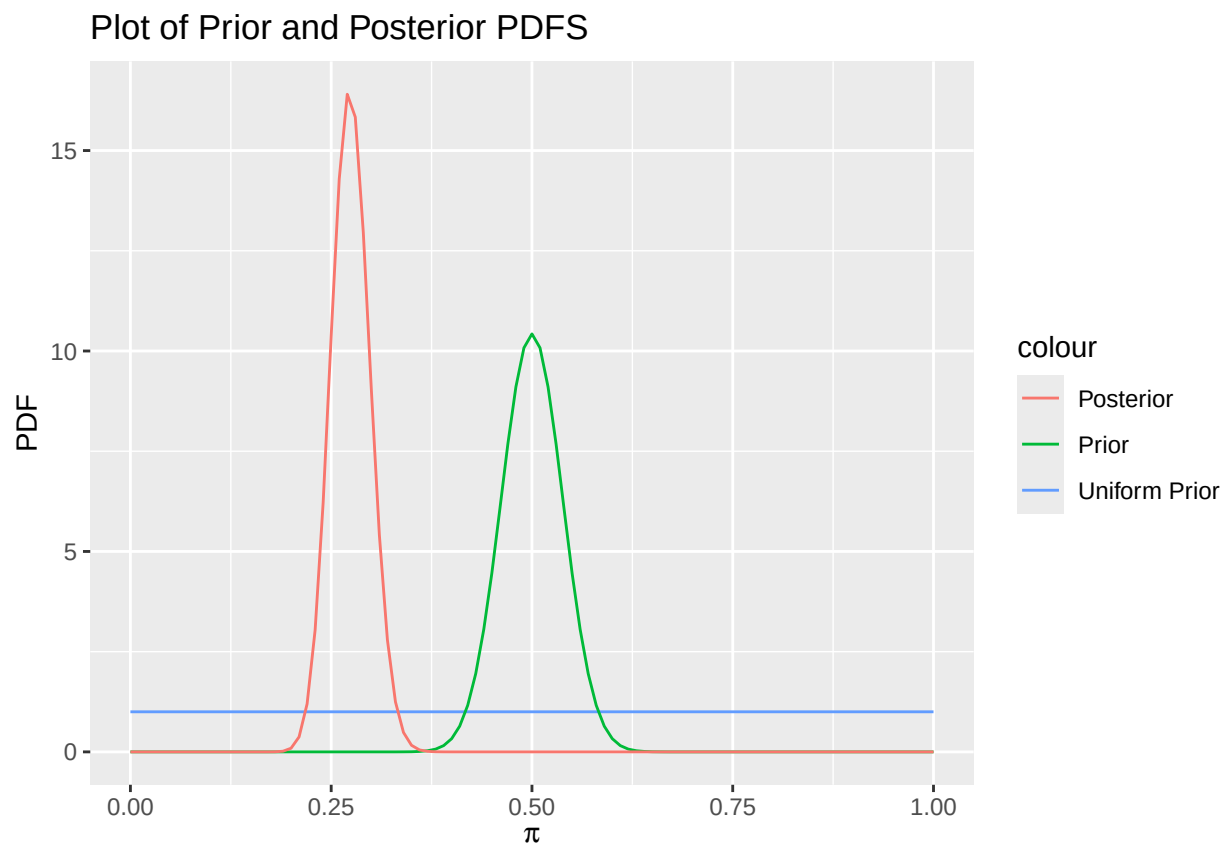
$$\pi|t \sim \text{Beta}(93.53, 247.53)$$

The distribution of this is:

```
ggplot() +
  geom_function(fun = dbeta,
    mapping = aes(color = "Uniform Prior"),
    args = list(shape1 = 1, 1),
    xlim = c(0, 1)) +
```

```
geom_function(fun = dbeta,
              mapping = aes(color = "Prior"),
              args = list(shape1 = 85.63, 85.63),
              xlim = c(0, 1)) +
geom_function(fun = dbeta,
              mapping = aes(color = "Posterior"),
              args = list(shape1 = 93.53, 247.53),
              xlim = c(0, 1)) +

labs(
  title = "Plot of Prior and Posterior PDFS",
  x = expression(pi),
  y = "PDF"
)
```



It is seen that the posterior is skewed in relation to the prior, so the Highest Posterior Density Interval can be also used to determine the interval.

The median of π_i of the posterior distribution is

```
posterior_median_pi <- qbeta(0.5, shape1 = 93.53, shape2 = 247.53)
round(posterior_median_pi, 4)
```

```
## [1] 0.2738
```

```
qbeta(p = c(0.025, 0.975), shape1 = 93.53, shape2 = 247.53)
```

```
## [1] 0.2282513 0.3227230
```

```
library(HDInterval)
```

```
## Warning: package 'HDInterval' was built under R version 4.4.3
```

```
hdi(qbeta, credMass = 0.95, shape1 = 93.53, shape2 = 247.53)
```

```
##      lower      upper
## 0.2274001 0.3218083
## attr(,"credMass")
## [1] 0.95
```

The value of the median is 0.2738 and 95% HPDI posterior interval for π are (0.2274, 0.3227). These values can be used to solve for the values of the median and interval for ψ . For the median:

$$P(\pi < 0.2738|t) = 0.5$$

$$P\left(\frac{1-\psi}{2-\psi} < 0.2738|t\right) = 0.5$$

$$P(1-\psi < 0.2738(2-\psi)|t) = 0.5$$

$$P(1-\psi < 0.5478 - 0.2738\psi|t) = 0.5$$

$$P(-0.7262\psi < -0.4522|t) = 0.5$$

$$P(\psi > 0.6227|t) = 0.5$$

So, the median of ψ is 0.6227.

For the interval: It is known that:

$$P(\pi < 0.2274|t) = 0.025$$

$$P\left(\frac{1-\psi}{2-\psi} < 0.2274|t\right) = 0.025$$

$$P(1-\psi < (2-\psi)0.2274|t) = 0.025$$

$$P(1-\psi < 0.4548 - 0.2274\psi|t) = 0.025$$

$$P(-0.7726\psi < -0.5452|t) = 0.025$$

$$P(\psi > 0.7057|t) = 0.025$$

It is also known that:

$$P(\pi < 0.3218|t) = 0.975$$

So:

$$P\left(\frac{1-\psi}{2-\psi} < 0.3218|t\right) = 0.975$$

$$P(1-\psi < (2-\psi)0.3218|t) = 0.975$$

$$P(1-\psi < 0.6436 - 0.3218\psi|t) = 0.975$$

$$P(-0.6782\psi < -0.3568|t) = 0.975$$

$$P(\psi > 0.5261|t) = 0.975$$

This means the highest posterior density interval for ψ , vaccine efficacy, is 52.61% to 70.57%.

The null hypothesis, as stated earlier is:

$$H_o : \psi = 0.3$$

with an alternative that vaccine efficacy is higher than 30%:

$$H_1 : \psi > 0.3$$

The posterior probability is represented by

$$P(\psi \leq 0.3|t)$$

In terms of π , since its distribution is known, this probability statement is:

$$P\left(\frac{1-2\pi}{1-\pi} \leq 0.3|t\right)$$

$$P(1-2\pi \leq (1-\pi)0.3|t)$$

$$P(1-2\pi \leq 0.3-0.3\pi|t)$$

$$P(-1.7\pi \leq -0.7|t)$$

$$P\left(\pi \geq \frac{7}{17}\right)$$

Since π has a beta distribution:

```
1 - pbeta((7 / 17), 93.53, 247.53)
```

```
## [1] 5.684824e-08
```

The probability of vaccine efficacy being less than or equal to 30% is incredibly small, which means the null hypothesis can be rejected since there is evidence that the vaccine efficacy is above 30%.